Average Rate of Change

Section 1: Formula and Secant Line Concept

1 State the formula for the average rate of change of $f(x)$ over $[a, b]$.

a. $\frac{f(b) - f(a)}{b - a}$

b. $\frac{f(a) + f(b)}{2}$

c. $\frac{f(b) - f(a)}{2}$

d. $\frac{b - a}{f(b) - f(a)}$

2 Find the average rate of change of $f(x) = x^2 - 3x$ from $x = 1$ to $x = 4$.

a. 2

b. 4

c. 3

d. 1

3 Find the slope of the secant line connecting $(2, f(2))$ and $(5, f(5))$ for $f(x) = 2x^2 + 1$.

a. 14

b. 12

c. 16

d. 10

Section 2: Tabular and Data-Based Rates of Change

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4 A table shows $t = [0, 2, 5]$ and distance $s(t) = [0, 24, 75]$. What is the average velocity from $t = 2$ to $t = 5$?

a. 17 m/s

b. 15 m/s

c. 25 m/s

d. 12 m/s

5 Water depth in a reservoir drops from 80 ft at day 4 to 68 ft at day 10. Find the average rate of change in depth per day.

a. -2 ft/day

b. 2 ft/day

c. -1.2 ft/day

d. -12 ft/day

6 If the average rate of change of $g(x)$ over $[1, 6]$ is 4 and $g(1) = 5$, find $g(6)$.

a. 25

b. 20

c. 29

d. 24

Section 3: Algebraic Expressions with Variables

7 Find the average rate of change of $f(x) = x^2$ over the interval $[2, 2 + h]$ in terms of h.

a. $4 + h$

b. 4

c. $2 + h$

d. $4 + 2h$

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8 Find the average rate of change of $f(x) = 3x^2 - 5$ over $[x, x + h]$.

a. $6x + 3h$

b. $6x$

c. $3x + 3h$

d. $6x - 5$

9 For $f(x) = \frac{1}{x}$, find the average rate of change over $[a, b]$.

a. $-\frac{1}{ab}$

b. $\frac{1}{ab}$

c. $-\frac{1}{a+b}$

d. $\frac{a-b}{ab}$

Section 4: Contextual Applications

10 A projectile's height is given by $h(t) = -16t^2 + 80t$. Find its average velocity over the first 3 seconds $[0, 3]$.

a. 32 ft/s

b. 48 ft/s

c. 16 ft/s

d. 80 ft/s

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11 A company's revenue in thousands of dollars is $R(x) = 50x - x^2$, where x is in hundreds of units. What is the average rate of change in revenue as production increases from $x = 10$ to $x = 20$?

a. 20 thousand dollars per unit

b. 30

c. 15

d. 25

12 Why does a linear function $f(x) = mx + b$ have the same average rate of change over EVERY interval?

a. Because the slope of the secant line between any two points is constantly m

b. Because $b = 0$

c. Because the function has no intercepts

d. Because $f(x)$ is symmetric

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